Multiscale–Sparse Spatial–Spectral Transformer for Hyperspectral Image Denoising

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***Abstract*— Improving hyperspectral image (HSI) quality is crucial in subsequent applications. Current transformer-based methods effectively remove mixed noise from the original HSIs. However, there is limited research on targeted modeling of the spatial locality and edge properties of HSI. In addition, the original transformer computes global query–key pairs indifferently, resulting in equal weights for dissimilarity features, noise, and essential information, thus interfering with the restoration of clean images. To address these issues, this study proposes an effective HSI denoising network called multiscale–sparse spatial–spectral transformer (MS3T) to achieve end-to-end mixed noise removal. Specifically, we reconstruct the attention module in the original transformer by adopting a dual-stream approach to selectively explore the 3-D information of HSI from both spatial and spectral perspectives. In the spatial domain, we construct a multiscale context-capturing module based on the local and non-local similarity properties of HSI to establish remote connections from local to global. In the spectral domain, we develop a top-k selection operator to calculate the similarity scores of query–key pairs and select important semantic information for efficient feature aggregation. Both of the above modules alleviate the computational complexity issue of the original transformer to different extents, and improve the mixed noise removal performance of the whole network. To validate the effectiveness and efficiency of MS3T, we conduct synthetic and real experiments on multiple datasets, and the final results demonstrate that our method outperforms the state-of-the-art methods in both metric evaluation and visual effects; the reproducible code is available at https://github.com/Zilong-Xiao/MS3T.**

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I. Introduction

HYPERSPECTRAL images (HSIs) possess the capability to accurately represent detailed spatial and spectral information, describing the spectral reflection and radiation characteristics of different substances. Therefore, HSIs play an indispensable role in various fields such as remote sensing [1], medicine [2], agriculture [3], and food [4]. However, when it comes to specific applications such as object detection [5], [6], [7], [8], classification [9], [10], unmixing [11], fusion [12], [13], and others [14], [15], [16], [17], performance often deteriorates due to the interference of image noise [18]. Constrained by imaging conditions and external interference, the imaging, acquisition, and transmission of images generate different types and intensities of noise in different bands, such as Gaussian noise, impulse noise, stripes, deadlines, etc. [19], [20], [21]. Although noise generation can be reduced through hardware optimization methods, they are costly and device-specific [22], [23]. Considering cost-effectiveness and versatility, HSI noise removal is an essential step in ensuring the efficiency of follow-up advanced applications.

Owing to the complexity of noise sources and types, the analysis and modeling of noise are highly challenging. Over the past decades, extensive studies have been conducted on HSI denoising, broadly categorized into two main approaches: model-based and learning-based methods. In model-based methods, researchers have incorporated suitable prior information, such as non-local similarity [24], [25], [26], sparse representation (SR) [27], [28], total variation (TV) [18], [27], [29], and low-rank [30], [31], [32], to transform the denoising problem from an ill-posed problem to a well-posed problem. These targeted handcrafted priors, which have clear physical significance, allow for precise modeling of image structure and feature information. However, deploying complex handcrafted priors entails constructing intricate objective functions, which not only lack generality but also bring parameter sensitivity to optimization. Furthermore, the optimization process consumes massive resources and yields unstable results.

In recent years, learning-based methods have flourished, especially methods based on convolutional neural networks (CNNs) [23], [33]. These methods abandon the complex handcrafted priors in model-driven approaches and instead

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utilize paired clean-noisy images to learn implicit priors [23], [33], [34], [35], [36], [37], [38]. They rely on less physical degradation models, which make them versatile and fast. However, CNN-based models are not suitable for processing sequential data due to their characteristics of local connections and weight sharing. The limited receptive field and fixed weights in the smallest unit (i.e., convolutional layer) result in weak robustness when dealing with noisy and deformed data. These limitations restrict the flexibility and denoising ability of CNN-based models on large-scale real-world datasets.

Researchers have introduced transformer into the field of computer vision to address various limitations of CNNs. Vision transformer (ViT) [39], compared to CNNs, can utilize attention mechanisms to capture global features while effectively modeling long-range dependencies. Moreover, due to the weak inductive bias of the transformer, it exhibits stronger learning ability in large amounts of data, addressing the receptive field and weight limitations of CNNs [20], [21], [40], [41], [42], [43], [44]. However, there remains a scarcity of appropriate transformer models for targeted HSI feature modeling. The limited local information extraction capability of the transformer restricts the exploration of spatial similarity in images, resulting in weak generalization ability. When calculating attention, the transformer directly aggregates features using all query–key pairs, which amplifies the impact of implicit noise with low similarity on information interaction. In addition, the computational complexity increases quadratically with the increase in resolution, leading to decreased efficiency.

Diffusion models, particularly self-supervised diffusion models [45], [46], [47], have become a new solution for HSI denoising due to their flexibility and generality. They do not require large amounts of paired datasets for training and can widely apply to various scenarios by capturing complex relationships and dependencies within data. However, there are also some challenges, the most obvious being the complexity of the model and the cumbersome inference process, which requires significant resources and time to process the image.

According to the above viewpoints, we propose a targeted HSI denoising residual network called multiscale–sparse spatial–spectral transformer (MS3T). The core module is the multiscale–sparse transformer block (MSTB) that includes dual-stream attention, which is used to model the spatial–spectral structure of HSIs and aggregate 3-D features. Specifically, in one branch of MSTB called multiscale multihead self-attention (MMSA), we first utilize different window partitioning methods to explore spatial contextual information of multiscale regions based on their spatial similarity characteristics. Then, we overlay the information to enhance local details and establish connections between global and local features, which can achieve precise noise removal while preserving background details. The other branch of MSTB is sparse multihead self-attention (SMSA), where we utilize a top-k attention mechanism to compute attention values along the spectral dimension to explore global correlations along the spectrum (GCS). In the computation, only the most valuable self-attention values are retained to mask implicit interference of noise on image restoration. The above designs bring three advantages in HSI denoising: 1) the ability to exploit local and global features in space has been enhanced; 2) the robustness against irrelevant features and complex noise has been enhanced; and 3) the partitioning of regions and selective computation of attention values can both reduce the computational complexity of the overall network.

The contributions of this work can be summed up as follows.

1) We propose a transformer-based residual network architecture, MS3T, to extract deep spatial–spectral feature representations for mapping residual noise and obtaining high-quality restoration results.
2) We introduce an effective MMSA module to explore spatial similarity, which greatly enhances modeling and information extraction capabilities.
3) We design a top-k operator to replace the original spectral attention calculation operation, which actively selects the most relevant attention values to achieve better feature aggregation.
4) The results of extensive experiments on removing a large amount of mixed noise, both real and synthetic, indicate that our method is superior to the state-of-the-art methods in terms of metrics and visual effects.

II. Related Work

A. Model-Based HSI Denoising

Considering the characteristic of numerous spectral bands in HSI. Therefore, early HSI denoising methods relied on strategies used in natural image denoising, such as the use of band-by-band denoising techniques such as non-local means (NLMs) filter [48], weighted nuclear norm minimization (WNNM) [49], and block matching and 3-D filter (BM3D) [25]. Nonetheless, these methods for single-band images ignore the correlation between adjacent bands in HSI. To maximize the utilization of rich spectral information, researchers have endeavored to improve existing algorithms. For instance, BM4D [26] groups the original 3-D data into 4-D data and mitigates residual low-energy noise through collaborative filtering. Moreover, model-driven approaches with handcrafted priors have become mainstream. According to the segmented smoothness property of HSI, introducing TV regularization to establish a prior term can effectively smooth random noise. On this basis, a spatial–spectral TV model is constructed simultaneously by adding spectral constraints. For example, Chang et al. [29] proposed an anisotropic spatial–spectral TV (ASSTV) framework to model HSI as a spatial–spectral volume to improve results smoothness. Peng et al. [50] proposed an enhanced 3-D TV (E3DTV) model, which calculates the sparsity of subspaces on the gradient maps of all bands to encode the original correlations and differences between bands. However, TV regularization loses texture during smoothing noise, so research on models based on SR has been conducted. Dabov et al. [51] used collaborative altering to group 2-D patches into a 3-D array for SR collaborative filtering. Zhang et al. [52] proposed group-based SR (GSR) to associate non-local patches and then perform sparse coding. Xu et al. [53] developed a trilateral weighted sparse coding

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(TWSC) strategy to describe hybrid noise and image priors. Nevertheless, due to the substantial rise in computational complexity resulting from the abundant spectral information, incorporating the low-rank properties of clean HSIs within the denoising model can effectively remove noise of varying types and intensities while consuming fewer computational resources. Specifically, this entails unfolding and organizing the 3-D HSI data into a 2-D matrix by dictionary order. Zhang et al. [30] proposed low-rank matrix recovery (LRMR) for removing mixed noise, while He et al. [54] proposed a unified framework of low-rank matrix and TV (LRTV). Furthermore, researchers have also developed a low-rank tensor model [55], [56], [57] that directly decomposes the original 3-D structure to preserve the low-rank attribute of HSI.

The advantage of model-based methods lies in their good interpretability and independence from data. However, in real-world random noisy environments, simple handcrafted priors are unable to achieve accurate estimation, and fixed models lack generalization for complex scenarios.

B. CNN-Based HSI Denoising

Learning-based methods (especially CNNs) achieve advanced denoising performance by directly establishing a nonlinear mapping relationship between clean images and noisy images through learning a large amount of data. During the initial exploration phase, CNN-based methods leverage experience from RGB image denoising to perform band-by-band noise removal. They extract feature information from HSI using 2-D convolutional filters. For instance, Chen and Pock [34] proposed a trainable nonlinear reaction–diffusion framework. Chang et al. [35] proposed HSI-DeNet, which is similar to the tensor-based model. However, these approaches disrupt the original spectral structure, hindering the full exploitation of the spectral correlation in HSIs. Consequently, the adoption of 3-D convolution [36], [38] with a robust spatial–spectral extraction capability has become prevalent. Meanwhile, to mitigate the additional computational cost associated with 3-D convolution, some methods [33], [58], [59] utilize a strategy that combines spatial 2-D convolution and spectral 1-D convolution as a substitute for 3-D convolution. Furthermore, to delve into the intrinsic spatial–spectral properties of HSIs and bolster denoising performance, researchers [60], [61] have integrated attention mechanisms into CNN models to capture the characteristics of HSIs. Unsupervised networks are also an effective denoising solution considering the difficulty in obtaining clean HSIs in practical environments. For instance, Nguyen et al. [62] proposed an unsupervised CNN using Stein's unbiased risk estimate (SURE) loss function for denoising HSIs with band-varying noise. Luo et al. [63] introduced the spatial–spectral constrained deep image prior (S2DIP) to exploit the inherent priors of clean and noisy HSIs for removing mixed noise. Miao et al. [64] utilized two unsupervised deep image priors to effectively model abundance maps and endmembers in HSIs. Nevertheless, the above models still lack a clear physical interpretation, and the complex data and uncertain outputs contribute to weak generalization and unstable performance. Given this, researchers have attempted to combine prior knowledge with deep networks to simultaneously leverage the inherent properties of HSIs and the powerful fitting capability of deep learning. For example, Wei et al. [38] introduced long-term spectral correlations into recursive neural networks. Zhang et al. [58] integrated the low-rank physical constraints with CNN to recover clean semantic information. Bodrito et al. [65] presented a trainable sparse spectral coding model (TS3C), which preserves the original sparse coding model and necessitates only a small amount of end-to-end data to train the network parameters. Xiong et al. [66] suggested the model-assisted non-local CNN (MAC-Net) by combining spectral low-rank prior and spatial depth prior. Fu et al. [67] established the non-local spatial–spectral neural network (NSSNN) based on the triple relationship of local spatial, global spectral, and spatial–spectral correlations.

Nevertheless, the limited receptive field and the local parameter-sharing mechanism of CNNs restrict the capability to model long-range dependencies, which is ineffective in exploiting spatial–spectral intrinsic similarity of HSIs.

C. Vision Transformer

The introduction of the transformer [68] has significant implications. Unlike RNN, which relies on sequences, and CNN, which focuses on local feature extraction, the transformer fully utilizes attention mechanisms to represent data with positional information, enabling effective learning of long-range dependencies in ordered data. Given the remarkable success of transformer in natural language processing (NLP), Dosovitskiy et al. [39] introduced the ViT, which converts images into sequences of patches to capture global relationships. This approach overcomes the limitations of sequential execution and the lack of parallel computation in RNNs, as well as the focus solely on local information in CNNs. Furthermore, the weak inductive bias of the transformer enables dynamic weight adjustment and feature prioritization, leading to significant performance improvements. Several efficient variants of transformer have been developed to handle various tasks flexibly. For example, the swin transformer [69] adopts a window partition strategy, enhancing local information extraction capability while reducing resource consumption. The pyramid vision transformer (PVT) [70] employs a gradually contracting pyramid pattern to reduce computational overhead in mapping, facilitating the execution of dense prediction tasks. Vision outlooker (VOLO) [42] achieves efficient feature extraction by encoding fine-grained features into tokens. In image restoration tasks, Wang et al. [71] introduced Uformer, a universal and efficient U-shaped transformer structure. SwinIR [72] integrates the swin transformer into a residual network as a backbone to explore the self-similarity within local windows. Moreover, to reduce the computational cost of high-resolution image restoration, Restormer [73] computes cross-covariance between channels to obtain feature maps. Considering the distinct characteristics of HSIs compared to RGB images, which possess richer spectral resolution and spatial–spectral relationships, most existing transformer-based methods incorporate

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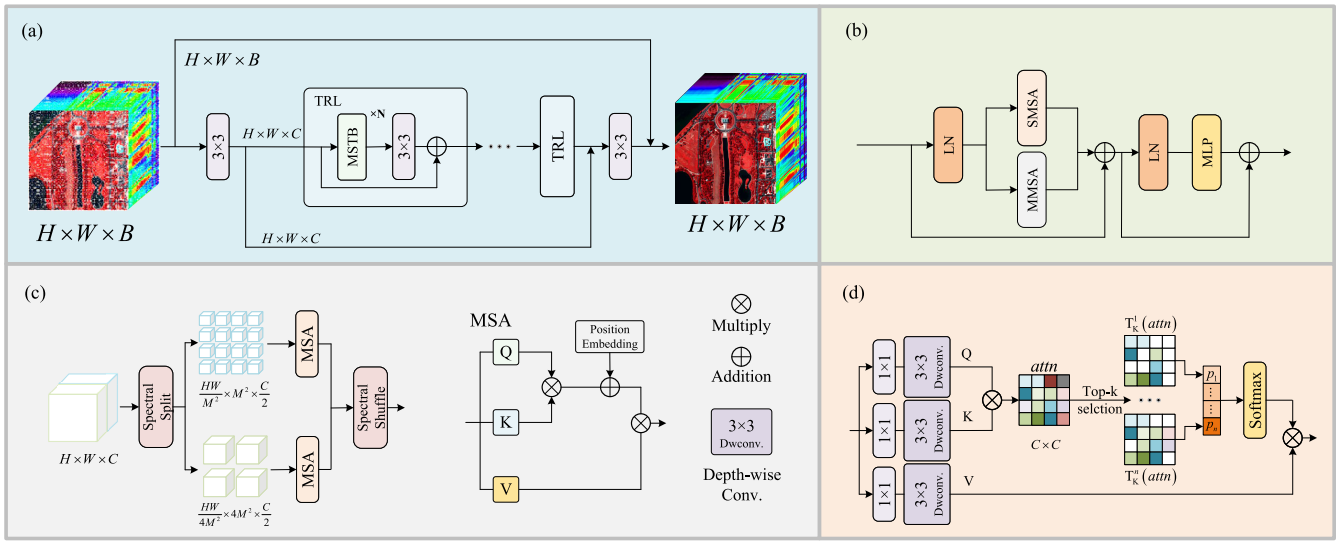


Fig. 1. Overall architecture of MS3T. (a) MS3T is a residual network architecture based on the transformer, where each residual layer (TRL) consists of multiple MSTBs. (b) Architecture of the MSTB. (c) Architecture of the MMSA. (d) Architecture of the SMSA.

dual spatial and spectral information constraints. For instance, the local–global feature-aware transformer (FATR) [20] incorporates feature maps obtained through spectral coarsening as inputs to the attention modules. HSI denoising transformer (Hider) [21] establishes the remote relationship between spectral and spatial aspects through an additional local enhancement module. Spatial–spectral transformer denoising (SSTD) [74] operates a spatial–spectral learning strategy to mitigate spectral distortion during the deep feature extraction. Moreover, the adoption of dual attention mechanisms to simultaneously explore spatial–spectral structural information has been widely employed, such as in spatial–spectral transformer (SST) [43], dual-stream transformer (DSTrans) [44], and other related methods. Concurrently, to address the limited local extraction capability of transformers, 3-D quasi-recurrent and transformer (TRQ3D) [75] combines the strengths of CNNs and transformers to perform denoising collaboratively. Spectral enhanced rectangle transformer (SERT) [76] simulates non-local spatial similarity using horizontal and vertical rectangular self-attention.

III. Methodology

In this section, we first overview the overall architecture of the transformer-based residual network MS3T, which is employed for removing hyperspectral mixed noise. Subsequently, we elaborate on our proposed attention blocks consisting of MMSA and SMSA.

A. Overall Network Architecture

For the degradation of HSIs, we can intuitively model it as

$$\boldsymbol{Y} = \boldsymbol{X} + \boldsymbol{N} \tag{1}$$

where $\boldsymbol{Y} \in \mathbb{R}^{H\times W\times B}$ and $\boldsymbol{X} \in \mathbb{R}^{H\times W\times B}$ represent the observed HSI and clean HSI, respectively, while $\boldsymbol{N} \in \mathbb{R}^{H\times W\times B}$ expresses the mixed noise. H, W, and B represent the height, width, and spectral channels of the HSI, respectively. Motivated by this intuitive physical modeling, we propose a residual network MS3T for end-to-end training, aiming to learn the implicit mapping relationship between paired noise and clean HSIs. Similar approaches have been validated in various low-level vision tasks, demonstrating improved generalization ability while avoiding overfitting.

The details of MS3T are illustrated in Fig. 1(a). The noisy HSI $\boldsymbol{I}_n \in \mathbb{R}^{H\times W\times B}$ is fed into the network. Initially, we employ a 3×3 convolution for feature embedding, transforming the HSI into a specific channel C, which serves as a shallow feature extraction step. Subsequently, we utilize six transformer-based residual layers (TRL) to extract deep-level features, where each layer contains six multiscale–sparse transformer blocks (MSTBs). The structure of MSTB is shown in Fig. 1(b); we construct two attention mechanisms to blend their effects. MMSA builds spatial local invariance and similarity, while SMSA utilizes the sparsity of HSIs to calculate correlation weights. The above mechanisms extract useful contextual information from both spatial and spectral domains and then merge them under the dual constraints of space and spectrum. The detailed description of MMSA and SMSA will be discussed in the following section. Additionally, we adopt layer normalization (LN) [77] to stabilize the distribution of intermediate layer inputs in the network and accelerate the convergence rate. A 3×3 convolutional layer is employed behind each TRL layer to enhance the extracted features. At the end of the entire network, the residual connections merge shallow geometric features with deep abstract contextual information, and then the reconstructed high-quality HSI is generated via convolutional layer mapping.

B. Multiscale Multihead Self-Attention

Exploring spatial similarity is crucial for HSI denoising. However, CNN-based methods are limited by the receptive field issue, as they fail to model the characteristics of HSI effectively. On the other hand, although original transformer networks can overcome long-range dependencies, they exhibit poor capability in extracting local information. Moreover,

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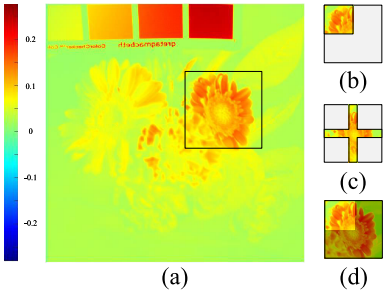


Fig. 2. (a) Heatmap of the flowers_ms image from the CAVE dataset is used to visually represent pixel similarities. (b) Window attention of the swin transformer. (c) Rectangle attention of SERT. (d) Multiscale attention of MS3T. The swin transformer emphasizes neighboring pixel similarity but often overlooks similarities among distant features, whereas SERT has lower feature utilization. Our MS3T excavates local information and constructs connections from local to remote.

their computational complexity grows quadratically with the increase in image size. Given the powerful performance of transformers, many studies have made improvements to transformers to adapt them to denoising tasks. For instance, the swin transformer [69] partitions the input image into small windows and enables inter-window information interaction through a sliding window mechanism. Meanwhile, SERT [76] divides the image into horizontal and vertical rectangles and exploits non-local similarity in HSIs. To visualize the similarity between pixels, we generate a heatmap by mapping the grayscale values of pixels, which can reflect the intensity, density, and activity level of the pixels. As demonstrated by the heatmap in Fig. 2(b) and (c), the swin transformer tends to focus on the similarity of adjacent pixels and overlooks remote similar features, while SERT has a relatively low utilization rate for features. To address this, we propose MMSA, which divides the feature map into blocks of varying sizes. The central area of each block is assigned a higher weight, allowing for greater attention to highly similar neighboring pixels while also taking remote similarities into account, as demonstrated in Fig. 2(d). In prior research [74], this multiscale feature extraction method using different block sizes has been demonstrated to be effective.

The details of MMSA are illustrated in Fig. 1(c). We employ spectral split to divide the input HSI into two parts. Each part is further divided into blocks of different spatial sizes, which serve as inputs for attention computation. Subsequently, the branch information of different scales is connected together through spectral shuffle to obtain comprehensive features. We represent the input features of MMSA as $\boldsymbol{Z} \in \mathbb{R}^{H\times W\times C}$. The overall process can be represented mathematically as follows:

$$\begin{aligned} \boldsymbol{Z}_1, \boldsymbol{Z}_2 &= \text{Split}(\boldsymbol{Z}) \\ \hat{\boldsymbol{Z}}_1 &= \text{L-MMSA}(\boldsymbol{Z}_1), \hat{\boldsymbol{Z}}_2 = \text{S-MMSA}(\boldsymbol{Z}_2) \\ \hat{\boldsymbol{Z}} &= \text{Shuffle}\left(\hat{\boldsymbol{Z}}_1, \hat{\boldsymbol{Z}}_2\right). \end{aligned} \tag{2}$$

Among them, $\boldsymbol{Z}_1$ and $\boldsymbol{Z}_2 \in \mathbb{R}^{H\times W\times (C/2)}$ are two parts divided by $\boldsymbol{Z}$, where the functions Split and Shuffle represent the spectral partitioning and rearrangement operations, respectively. L-MMSA and S-MMSA denote attention operations at large and small scales, respectively. For L-MMSA, the feature map $\boldsymbol{Z}_1$ is divided into $(HW/4M^2)$ sub-blocks of size $4M^2 \times (C/2)$, denoted as $\{\boldsymbol{Z}_1^1, \boldsymbol{Z}_1^2, \ldots, \boldsymbol{Z}_1^{(HW/4M^2)}\}$. The attention operation for each sub-block is defined by the following equation:

$$\begin{aligned} \boldsymbol{Q}_1^i = \boldsymbol{Z}_1^i \boldsymbol{W}_1^Q, \boldsymbol{K}_1^i = \boldsymbol{Z}_1^i \boldsymbol{W}_1^K, \boldsymbol{V}_1^i = \boldsymbol{Z}_1^i \boldsymbol{W}_1^V \\ \hat{\boldsymbol{Z}}_1^i = \text{SoftMax}\left(\frac{\boldsymbol{Q}_1^i \boldsymbol{K}_1^{i\top}}{\sqrt{d}} + \boldsymbol{B}\right)\boldsymbol{V}_1^i \end{aligned} \tag{3}$$

where $\boldsymbol{W}_1^Q, \boldsymbol{W}_1^K$, and $\boldsymbol{W}_1^V \in \mathbb{R}^{(C/2)\times(C/2)}$ represent the projection mappings for the $\boldsymbol{Q}_1^i, \boldsymbol{K}_1^i$, and $\boldsymbol{V}_1^i \in \mathbb{R}^{4M^2\times(C/2)}$, respectively. The parameter d denotes the feature dimension, while $\boldsymbol{B}$ is a learnable positional embedding parameter, derived from the bias matrix $\hat{\boldsymbol{B}} \in \mathbb{R}^{(2M-1)\times(2M+1)}$. The SoftMax function can constrain the output values within a fixed range, thereby enhancing the stability of network optimization. For S-MMSA, the calculation method is quite similar to L-MMSA, with the key difference being that the feature map $\boldsymbol{Z}_2$ is divided into (HW/M^2) sub-blocks of size $M^2 \times (C/2)$ when entering the MSA.

C. Sparse Multihead Self-Attention

Many studies have proposed spectral attention-based methods to exploit the spectral correlations in HSIs for image restoration and fidelity enhancement. However, standard transformers often aggregate similar features based on global query–key pairs, leading to interactions between irrelevant information that can interfere with the restoration results. To overcome this detrimental effect, we propose SMSA as a replacement for the original spectral attention mechanism. By scoring the similarity of features, SMSA adaptively preserves relevant components while discarding irrelevant ones, thus avoiding interference from redundant information during information interaction.

Specifically, to address spectral attention, we transpose the input $\boldsymbol{Z} \in \mathbb{R}^{H\times W\times C}$ into $\boldsymbol{Z}' \in \mathbb{R}^{C\times H\times W}$ and employ 1×1 convolution and 3×3 depth-wise separable convolution to encode the original information. The workflow of SMSA is illustrated by the following equation:

$$\text{SMSA}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \text{SoftMax}\left(\sum_{m=1}^{n}\left(\text{T}_K^m\left(\frac{\boldsymbol{Q}\boldsymbol{K}^\top}{\sqrt{d}}\right) \times p_m\right)\right)\boldsymbol{V} \tag{4}$$

where the calculation of $\boldsymbol{Q}, \boldsymbol{K}$, and $\boldsymbol{V} \in \mathbb{R}^{C\times HW}$ follows (3). The d is the feature dimension, while the computed value of the query–key pair represents the global similarity score. Inspired by the top-k mechanism [78], [79], [80], [81], we filter the obtained attention matrix of size $\mathbb{R}^{C\times C}$ selectively through n top-k selection operators $\{\text{T}_K^1(\cdot), \text{T}_K^2(\cdot), \ldots, \text{T}_K^n(\cdot)\}$. Correspondingly, n learnable parameters $\{p_1, p_2, \ldots, p_n\}$ are employed to adaptively adjust the proportional contribution of each operator's value. A top-k operator can be defined as follows:

$$\begin{aligned} \text{T}_K^m(\text{attn}) &= \begin{cases} \text{attn}_{ij}, & \text{attn}_{ij} > \theta \\ -\infty, & \text{otherwise} \end{cases} \\ \theta &= (1-k)\max(\text{attn}_i) + k \times \min(\text{attn}_i) \end{aligned} \tag{5}$$

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where attn $\in \mathbb{R}^{C\times C}$ represents the calculated values of query–key pairs, indices i and j denote the row and column, and k is a variable parameter within the range $[\Delta_1, \Delta_2]$ that controls the sparsity of the matrix. Unlike the original top-k mechanism where only the first k values per row are retained, we calculate a threshold θ using the maximum and minimum values of each row and retain values above it. When data distribution is uneven, selecting a fixed k number of values still includes a lot of irrelevant information that causes interference. Using a threshold to filter out portions with low similarity scores can effectively eliminate this redundancy. Additionally, we vary the value of k to perform multiple $\mathrm{T}_K(\cdot)$ operations, aiming to adaptively explore both strongly and weakly correlated contextual information to achieve attention from dense to sparse. Finally, we multiply the softmax results with values using matrix multiplication and concatenate the attention outputs from all heads. The final result is obtained through linear mapping.

IV. Experiments

A. Experiment Setup

1) Datasets: We choose the CAVE dataset [82] for training and validation. This dataset comprises 32 HSIs, each with dimensions of 512 × 512 × 31. We select 26 HSIs and divide them into smaller sub-images measuring (64, 64, 31) using a stride of (16, 16). To enhance the diversity of our dataset, we utilize data augmentation techniques following [83]. As a result, we generate approximately 50 000 training samples. One-half of the remaining six HSIs are used for validation, while the others are used for testing, both without cropping and resizing. To evaluate the effectiveness of our method across diverse scenarios, we conduct synthetic experiments using the remote sensing datasets Chikusei [84] and Pavia University, and real-data experiments using Urban, Cuprite, Indian Pines, and the River. To tackle the issue of the smaller number of HSI data for fine-tuning and to assess the robustness of our method, we utilize the CAVE dataset-trained model to evaluate all the data. Moreover, considering the disparity in the band and size of the HSIs, we partition them into sub-images to facilitate a comprehensive comparison of denoising performance during testing. Descriptions of each dataset are as follows.

1) *CAVE*: It covers a range of real-world objects, including paintings, foods, toys, and fur, with a resolution of 512 × 512 × 31. Covering spectral bands from 400 to 700 nm, this dataset is utilized for training and conducting synthetic experiments.
2) *Chikusei*: This dataset encompasses agricultural and urban areas of Chikusei, Ibaraki, Japan. It includes 128 spectral bands spanning from 363 to 1018 nm, with a spatial resolution of 2517 × 2335 pixels. For synthetic experiments, the dataset is partitioned into sub-images of size 512 × 512 × 31.
3) *Pavia University*: The dataset is collected over Pavia, Italy, and comprises 610 × 340 pixels with 115 bands. For synthetic experiments, the dataset is partitioned into sub-images of size 256 × 256 × 31.
4) *Urban*: The dataset consists of land cover classes, including road, roof, grass, and tree, with image dimensions of 307 × 307 and covering 210 spectral bands from 400 to 2500 nm. We partition it into sub-images of size 256 × 256 × 31 for real-data experiments.
5) *Cuprite*: It comprises Cuprite in Las Vegas, NV, U.S., with image dimensions of 512 × 614 and covering 224 spectral bands from 370 to 2480 nm. For real-data experiments, we segment it into sub-images of size 512 × 512 × 31.
6) *Indian Pines*: This dataset primarily incorporates three regions: agriculture, vegetation, and forest. The images embrace 145 × 145 pixels with 224 bands. For real-data experiments, we crop it into sub-images of size 128 × 128 × 31.
7) *The River*: This dataset is collected from a river area in Jiangsu Province, China, using the EO-1 Hyperion sensor with a spatial resolution of 463 × 241 and consisting of 242 spectral bands. For real-data experiments, we partition it into sub-blocks, each containing 31 bands.

2) Comparison Algorithms: In both synthetic and real-data experiments, 10 algorithms, including model-based and learning-based methods, are selected for comprehensive comparison with our proposed MS3T. These include the following: 1) three tensor-based low-rank methods, namely, LLRT [55], LRTDTV [56], and NGmeet [57]; 2) two CNN-based methods, namely, QRNN3D [38] and NSSNN [67]; 3) a hybrid model-based and CNN-based method, namely, MAC-Net [66]; 4) a deep-unfolding based method, namely, SMDS-Net [85]; and 5) three transformer-based methods, namely, TRQ3D [75], SST [43], and SERT [76].

3) Noise Settings: To better simulate real-world noise and thoroughly compare the denoising performance of various methods, we superimpose i.i.d. and non-i.i.d. Gaussian, deadline, stripe, and impulse noise on the clean HSI for training and testing. The six cases of noise situations are introduced as follows:

1) *Case 1 (i.i.d. Gaussian Noise)*: All bands of the HSI cube are simulated with zero-mean Gaussian noise of the same standard deviation. To ensure fairness in the experiment, four sub-cases, case1.1∼1.4, are set to simulate Gaussian noise with standard deviations of {10, 30, 50, 70}, respectively.
2) *Case 2 (Non-i.i.d. Gaussian Noise)*: Given that real-world HSIs are often affected by complex environmental factors, we simulate non-i.i.d. Gaussian noise. Each band is randomly simulated zero-mean Gaussian noise with standard deviations ranging from 10 to 70.
3) *Case 3 (Non-i.i.d. Gaussian Noise + Deadline Noise)*: Based on Case 2, one-third of the bands are picked at random and have deadline noise added, with the column length ranging from 5% to 15%.
4) *Case 4 (Non-i.i.d. Gaussian Noise + Stripe Noise)*: The simulation method for stripe noise is the same as that for deadline noise in Case 3.
5) *Case 5 (Non-i.i.d. Gaussian Noise + Impulse Noise)*: Based on Case 2, one-third of the bands are randomly

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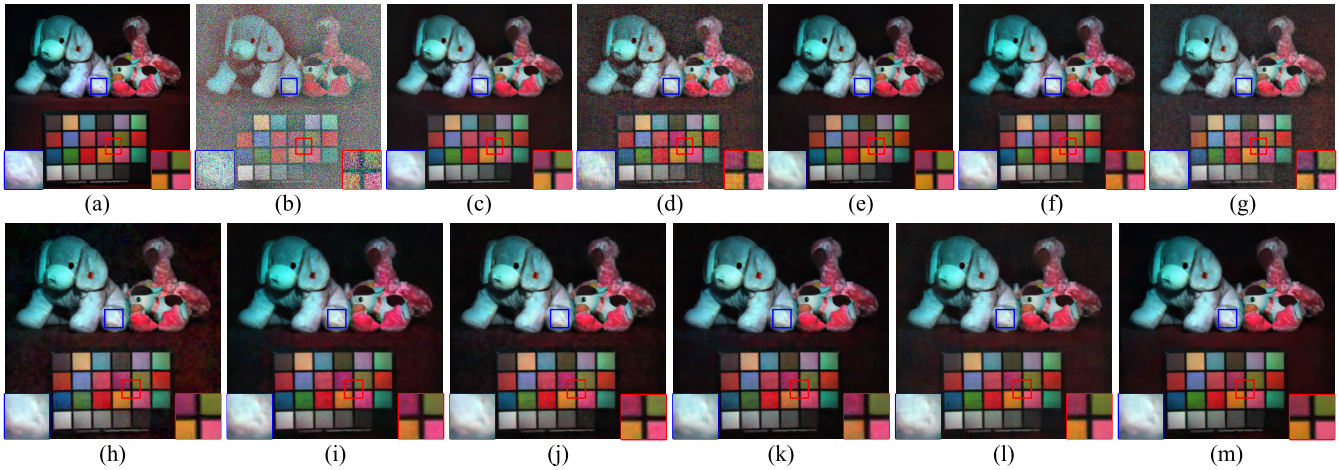


Fig. 3. Qualitative comparison of the CAVE dataset under Case 1.3 setting, using bands (31, 11, 6) for synthesizing pseudo-color images. (a) Original image, (b) noisy image, (c) LLRT, (d) LRTDTV, (e) NGmeet, (f) QRNN3D, (g) MAC-Net, (h) SMDS, (i) TRQ3D, (j) NSSNN, (k) SST, (l) SERT, and (m) MS3T.

selected to simulate impulse noise with intensity ranging from 10% to 70%.

6) *Case 6 (Mixed Noise)*: Combine the noise setting of Case 2∼5 to simulate mixed noise.

4) Evaluation Metrics: Based on the 3-D characteristics of HSI, we select four metrics, namely peak signal-to-noise ratio (PSNR), structural similarity index (SSIM) [86], spectral angle mapper (SAM) [87], and relative dimensionless global error in synthesis (ERGAS) [88], to quantitatively assess the performance of all approaches. These metrics comprehensively consider factors such as spatial and spectral resolution of HSI and measure the distortion and similarity between the reconstructed and the original HSI. Among them, higher values of PSNR and SSIM and lower values of SAM and ERGAS represent better denoising performance.

B. Training Details

1) Loss Function: To optimize our network, the mean squared error (MSE) is used to quantify the difference between denoised HSI and ground-truth, which can be denoted as

$$\mathcal{L} = \frac{1}{N}\sum_{i=1}^{N} \|\mathrm{MS3T}(\boldsymbol{Y}_i) - \boldsymbol{X}_i\|_F^2 \tag{6}$$

where $\boldsymbol{Y}_i$ and $\boldsymbol{X}_i$ denote the pair of noisy and clean HSI, respectively. N represents the number of training patches in a batch.

2) Implementation Details: We train and test learning-based methods on a platform using the mainstream deep learning framework PyTorch 1.9.0, with an Intel Xeon E5-2650 v4 processor, 64GB memory, and NVIDIA RTX 1080Ti GPU. We use a platform equipped with an Intel Core i5-8400 CPU and 16GB memory to optimize and test model-based methods. Considering the complexity of real-world noise, we construct the training sets following two methods, Case 2 and Case 6, targeting pure Gaussian noise and mixed noise HSI, respectively. During training, we employ the Adam optimizer [89] for 100 epochs. The initial learning rate is set to 1×10^{-4}, which is divided by 10 after the 60th epoch. When performing feature embedding, the feature channel C is set to 90. In the six TRLs, each layer comprises six MSTBs while the number of attention heads is six. In Section III-B, M is set to 8, indicating that small and large windows in MMSA are of size 8 and 16, respectively. In SMSA, there are four top-k operators with k values of {1/2, 2/3, 3/4, 4/5}, and the initial value of the learnable parameter p is set to {0.2, 0.2, 0.2, 0.2}.

C. Synthetic Experiments

In this section, synthetic experiments are performed on three datasets, namely CAVE, Chikusei, and Pavia University, to assess the robustness of our method in denoising tasks across diverse scenarios.

1) CAVE Dataset: Figs. 3 and 4 present the denoising results of the CAVE dataset under Case 1.3 and Case 6, respectively. To better evaluate spatial details and spectral fidelity, the results are processed using pseudo-color rendering.

For Case 1.3, LRTDTV, MAC-Net, and SMDS retain some residual noise and artifacts. Thanks to the preservation of the 3-D characteristics of HSI by low-rank tensors; LLRT and NGmeet remove most of the noise but lose details. Given the multidimensional feature representation capability of 3-D CNN, QRNN3D effectively preserves image textures and restores spectral information. However, some residual artifacts are still visible upon zooming in. TRQ3D, NSSNN, and SST effectively utilize global spatial context, achieving good visual results. However, the lack of effective interaction between global and local information results in residual noise in local regions. By introducing the global low-rank property, SERT effectively suppresses noise. Additionally, benefiting from non-local similarity, it preserves most of the features, although some artifacts remain. Our approach integrates both local and non-local similarity in the spatial domain and enhances features by using sparse masking to suppress noise interference. During the restoration process, we simultaneously achieve noise removal and detail preservation.

For Case 6, the result of LRTDTV still contains Gaussian noise and structural noise. LLRT and NGmeet effectively utilize low-rank properties for noise suppression, yielding

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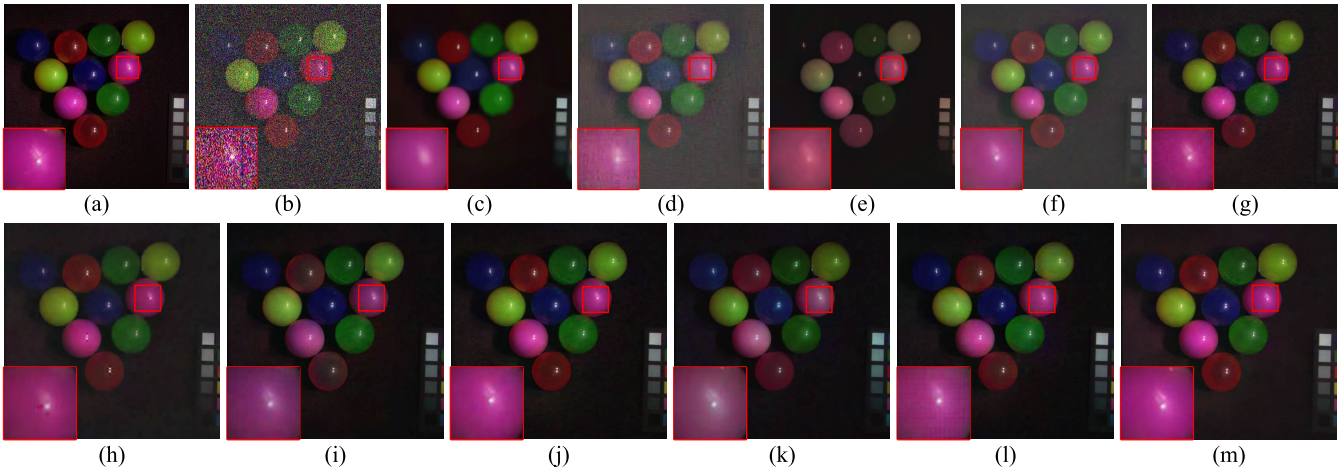


Fig. 4. Qualitative comparison of the CAVE dataset under Case 6 setting, using bands (30, 10, 4) for synthesizing pseudo-color images. (a) Original image, (b) noisy image, (c) LLRT, (d) LRTDTV, (e) NGmeet, (f) QRNN3D, (g) MAC-Net, (h) SMDS, (i) TRQ3D, (j) NSSNN, (k) SST, (l) SERT, and (m) MS3T.

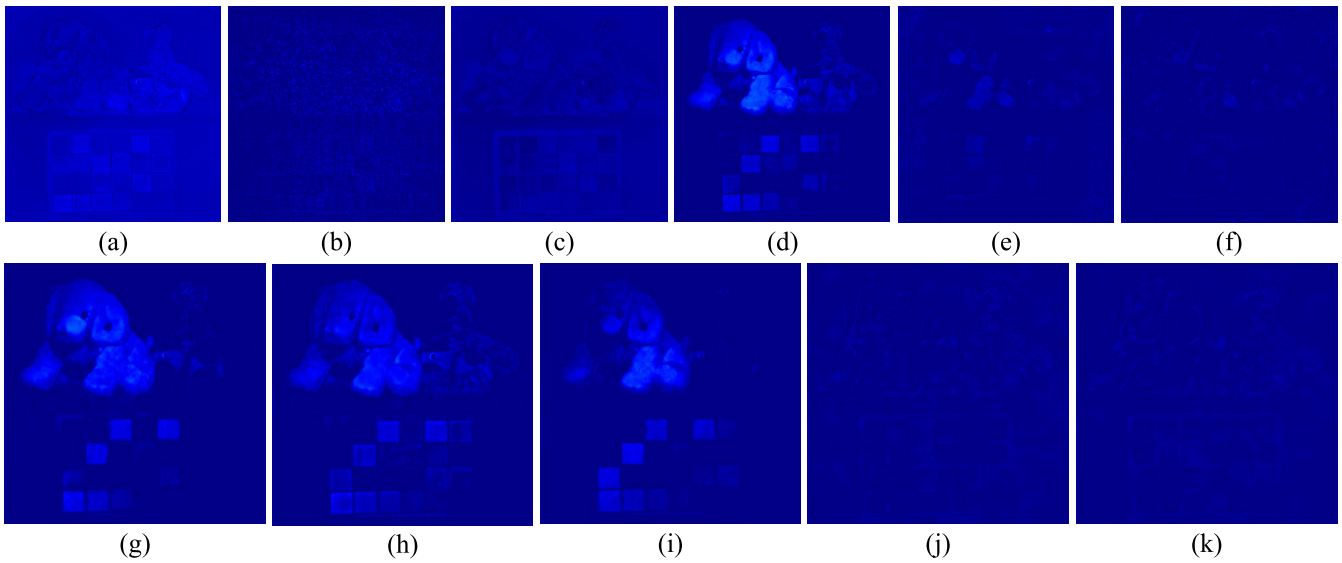


Fig. 5. Error maps of the CAVE dataset under Case 1.3 setting. (a) LLRT, (b) LRTDTV, (c) NGmeet, (d) QRNN3D, (e) MAC-Net, (f) SMDS, (g) TRQ3D, (h) NSSNN, (i) SST, (j) SERT, and (k) MS3T.

relatively clear results. However, the additional smoothing attributes lead to image blurring. Thanks to the effective modeling of HSI characteristics through structural spatial–spectral correlation, QRNN3D successfully preserves the details of spatial–spectral information while maintaining some residual noise. Unlike pure network-based approaches, the hybrid model MAC-Net and the deep unfolding model SMDS have clear physical significance and are specifically designed for modeling HSI. By observing the pseudo-color images, both methods exhibit remarkable performance in spectral restoration. However, the result of MAC-Net still exhibits noise, while SMDS suffers from excessive optimization, leading to over-smoothness. TRQ3D lacks the utilization of HSI spectral attributes, resulting in noticeable deviations in the pseudo-color images. SST leverages non-local similarity in the spatial domain to establish long-range connections for denoising but causes a loss of background information. In comparison to SST, SERT improves the attention mechanism in the spatial domain by effectively combining local and non-local features, yet some artifacts remain. Our method and NSSNN demonstrate superior performance by striking a balance between preserving spatial–spectral details and removing noise.

For a more visual indication of the denoising effectiveness, we illustrate the difference between the denoised HSI and the original HSI for Case 1.3 and Case 6 in Figs. 5 and 6, respectively. A cleaner error map indicates a closer resemblance to the original HSI. Our method and SMDS achieve the best restoration results, exhibiting almost no residue. Error analysis indicates that SMDS exhibits blocky artifacts, suggesting a slight weakness in spatial texture handling. In comparison to SMDS, our method excels in texture restoration, but with more pronounced errors along the edges. Additionally, to evaluate the extent of spectral distortion introduced by each method, we select the pixel location (150, 200) of "superballs_ms" HSI from the CAVE dataset under the settings of Case 6 to

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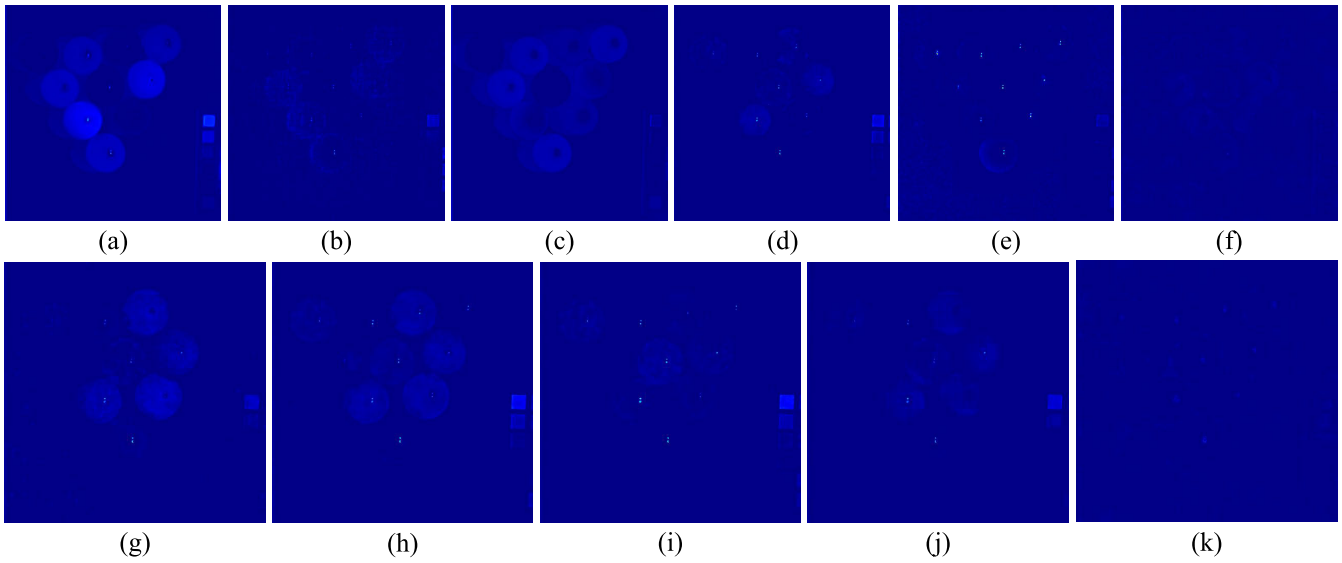


Fig. 6. Error maps of the CAVE dataset under Case 6 setting. (a) LLRT, (b) LRTDTV, (c) NGmeet, (d) QRNN3D, (e) MAC-Net, (f) SMDS, (g) TRQ3D, (h) NSSNN, (i) SST, (j) SERT, and (k) MS3T.

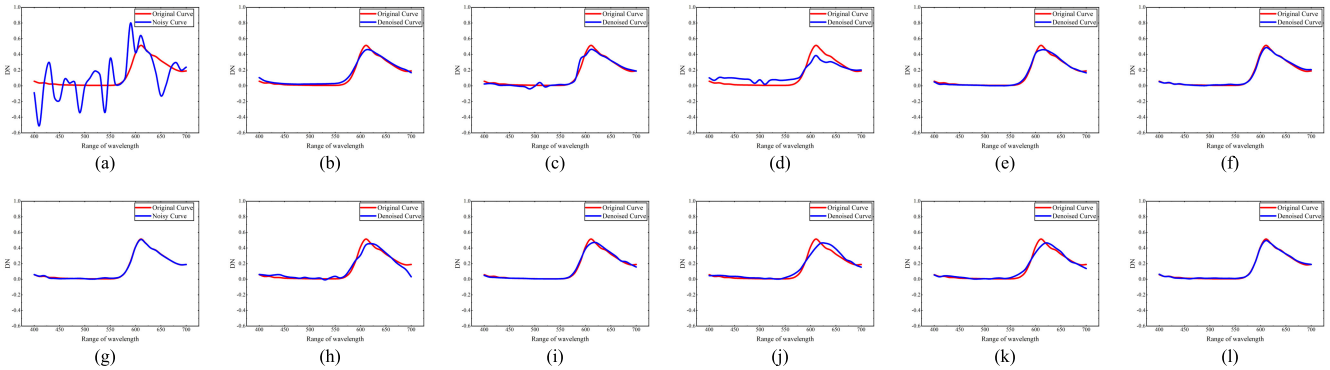


Fig. 7. DN value curves at pixel (150, 200) in the denoised images. The experiment is conducted on Case 6 of the CAVE dataset. (a) Noisy image, (b) LLRT, (c) LRTDTV, (d) NGmeet, (e) QRNN3D, (f) MAC-Net, (g) SMDS, (h) TRQ3D, (i) NSSNN, (j) SST, (k) SERT, and (l) MS3T.

plot the spectral restoration curves in Fig. 7, where the x-axis represents the spectral bands and the y-axis denotes the digital number (DN) values. From the graph, it can be observed that both our method and SMDS demonstrate superior spectral restoration performance.

In addition to the qualitative assessment of various methods, Table I presents a quantitative comparison of denoising effectiveness. In this table and all subsequent tables, we use red, green, and blue to stand for the first three indicators. These data are corroborated by actual visual effects. For low-intensity Gaussian noise, model-based methods achieve comparable results to deep learning methods due to their deterministic physical interpretation and better model optimization. However, when complex structural noise is introduced, deep learning methods demonstrate their advantage of being data-driven and performing better. The proposed method achieves state-of-the-art results in all metrics due to its targeted network design for spatial–spectral structures and the suppression of noise interference using sparse attention. Furthermore, Fig. 8 presents line graphs of PSNR and SSIM across different spectral bands in Case 1.3 and Case 6 to validate the consistency of denoising across spectra. Fig. 9 shows the convergence curves of training and validation in the simulation experiments. Our method exhibits high stability and robustness. In Table II, we further compare the performance of competing methods in runtime and computational complexity. Specifically, we process all HSIs at once and calculate the average processing time. The model-based approaches exhibit longer runtime due to factors such as running on CPUs and complex optimizations. Similarly, the SMDS method employs a deep unfolding strategy that requires intricate iterative optimizations, resulting in the longest runtime among the learning-based methods. Nevertheless, it has a simple network architecture and an extremely small number of parameters. When dealing with larger HSIs, the transformer model demonstrates faster inference speed due to its superior parallel computing capability. Consequently, transformer-based methods incur shorter runtime. Nonetheless, the complex structure of the transformer also results in a higher computational burden. Our method achieves notable performance improvement while maintaining a reasonable runtime and computational complexity.

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TABLE I
COMPARISON OF DENOISING METRICS FOR DIFFERENT METHODS ON THE CAVE DATASET

Case	Index	Noisy	LLRT	LRTDTV	NGmeet	QRNN3D	MAC-Net	SMDS	TRQ3D	NSSNN	SST	SERT	MS3T
i.i.d. Gaussian Noise													
Case 1.1	PSNR	28.13	42.16	43.40	46.40	40.49	39.12	46.00	37.69	39.26	38.47	44.61	46.69
	SSIM	0.7834	0.9931	0.9894	0.9960	0.9955	0.9692	0.9919	0.9857	0.9800	0.9958	0.9994	0.9996
	SAM	0.7756	0.1539	0.1545	0.0773	0.2037	0.2196	0.1249	0.1888	0.1649	0.1791	0.1213	0.0848
	ERGAS	185.63	23.74	33.71	22.95	67.22	36.50	24.30	80.80	58.71	67.93	28.77	22.53
Case 1.2	PSNR	18.59	39.98	37.26	41.29	38.61	36.09	41.49	36.73	37.37	36.91	41.65	43.02
	SSIM	0.3560	0.9841	0.9320	0.9848	0.9947	0.9190	0.9747	0.9802	0.9678	0.9944	0.9988	0.9991
	SAM	1.0463	0.1948	0.3629	0.1306	0.2409	0.3180	0.18845	0.2152	0.1929	0.2363	0.1602	0.1107
	ERGAS	557.07	50.87	65.67	41.47	74.48	66.78	40.66	85.83	67.94	79.55	39.84	33.96
Case 1.3	PSNR	14.15	38.93	32.96	38.71	37.14	35.68	38.19	35.80	36.55	35.70	39.38	40.75
	SSIM	0.2031	0.9747	0.8350	0.9737	0.9937	0.8367	0.9483	0.9712	0.9562	0.9928	0.9980	0.9985
	SAM	1.1617	0.1766	0.5589	0.1602	0.2880	0.4383	0.2680	0.2630	0.2185	0.2688	0.2084	0.1414
	ERGAS	928.31	55.26	107.33	55.57	82.60	87.35	59.24	93.21	75.96	90.47	51.35	43.98
Case 1.4	PSNR	11.23	37.23	29.84	34.65	35.79	33.33	36.13	34.92	35.84	34.68	37.45	38.97
	SSIM	0.1331	0.9163	0.7047	0.9375	0.9924	0.7452	0.9068	0.9616	0.9459	0.9912	0.9969	0.9978
	SAM	1.2341	0.3145	0.6943	0.2515	0.3494	0.5325	0.3757	0.3060	0.2421	0.2901	0.2690	0.1761
	ERGAS	1300.08	115.69	153.54	88.11	91.42	109.04	74.81	101.19	83.62	100.39	63.81	53.80
Non-i.i.d. Gaussian Noise													
Case 2	PSNR	16.49	34.93	33.13	32.62	38.57	38.65	39.11	35.81	40.04	35.89	41.67	43.30
	SSIM	0.2136	0.9304	0.7759	0.8806	0.9766	0.9445	0.9534	0.9463	0.9643	0.9616	0.9933	0.9956
	SAM	1.2977	0.2316	0.6903	0.3569	0.3444	0.3098	0.2256	0.3585	0.2161	0.3798	0.1899	0.1510
	ERGAS	1455.02	133.03	217.03	199.52	99.25	89.06	83.76	170.93	81.61	144.35	61.67	51.41
Non-i.i.d. Gaussian Noise + Deadline Noise													
Case 3	PSNR	18.41	35.27	33.15	32.04	39.49	40.46	39.34	35.99	40.43	35.87	42.21	43.64
	SSIM	0.3010	0.9245	0.7813	0.8603	0.9820	0.9637	0.9591	0.9460	0.9699	0.9595	0.9895	0.9957
	SAM	1.2819	0.2307	0.7458	0.5133	0.2833	0.2543	0.2168	0.3513	0.2269	0.3529	0.1787	0.1529
	ERGAS	1361.46	143.02	233.72	210.41	87.56	73.11	81.85	177.47	83.93	153.06	65.69	49.55
Non-i.i.d. Gaussian Noise + Stripe Noise													
Case 4	PSNR	18.36	33.27	33.96	32.02	39.01	40.01	39.38	35.51	40.08	35.17	41.73	43.78
	SSIM	0.2983	0.9236	0.8001	0.8507	0.9757	0.9609	0.9595	0.9553	0.9715	0.9587	0.9868	0.9958
	SAM	1.2814	0.2725	0.6905	0.5466	0.3410	0.2604	0.2150	0.3154	0.2074	0.3576	0.1728	0.1470
	ERGAS	1360.85	152.93	212.61	208.93	104.17	77.41	81.46	156.81	84.42	148.00	62.83	48.74
Non-i.i.d. Gaussian Noise + Impulse Noise													
Case 5	PSNR	18.96	34.98	33.99	32.68	39.32	39.82	39.35	35.85	40.46	35.95	42.17	43.87
	SSIM	0.3228	0.9454	0.8017	0.8869	0.9756	0.9682	0.9596	0.9493	0.9483	0.9617	0.9884	0.9957
	SAM	1.2744	0.2218	0.6866	0.3417	0.3594	0.2873	0.2132	0.3751	0.2142	0.3632	0.1779	0.1506
	ERGAS	1332.01	134.72	215.81	197.32	103.18	82.95	82.57	171.49	89.29	145.08	65.59	48.29
Mixed Noise													
Case 6	PSNR	19.43	35.12	34.76	32.04	39.99	40.21	39.57	36.03	41.15	35.79	42.23	43.98
	SSIM	0.3343	0.9344	0.8330	0.8666	0.9808	0.9638	0.9631	0.9583	0.9755	0.9643	0.9871	0.9959
	SAM	1.2589	0.2159	0.6388	0.5196	0.2827	0.2707	0.2100	0.2834	0.2121	0.3210	0.1904	0.1459
	ERGAS	1210.42	138.21	204.34	213.97	91.97	74.59	79.86	158.02	74.48	143.71	63.68	47.47

2) Chikusei Dataset: Compared to the CAVE dataset at close range, the Chikusei dataset has less spatial texture, and most methods achieve good denoising results, as shown in Fig. 10. However, the outcomes yielded by the LLRT approach tend to exhibit an over-smoothing effect after the denoising process. SMDS produces blurry effects in densely built areas. The visual effects of the remaining methods are relatively similar. As a supplement, the metrics in Table III more

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TABLE II
COMPARISON OF PARAMETERS AND TIME CONSUMPTION OF DIFFERENT METHODS IN CASE 6 OF CAVE DATASET

Index	Model-based (CPU)			Learning-based (GPU))							
	LLRT	LRTDTV	NGmeet	QRNN3D	MAC-Net	SMDS	TRQ3D	NSSNN	SST	SERT	MS3T
Params(M)	/	/	/	0.83	0.43	0.005	0.68	1.78	4.14	1.91	2.98
Time(s)	2921	99	244	12.5	22.67	620	15.09	14.53	7.41	5.17	5.74

TABLE III
COMPARISON OF DENOISING METRICS FOR DIFFERENT METHODS ON CASE 6 OF THE CHIKUSEI DATASET

Index	Noisy	LLRT	LRTDTV	NGmeet	QRNN3D	MAC-Net	SMDS	TRQ3D	NSSNN	SST	SERT	MS3T
PSNR	16.73	32.28	31.98	33.43	35.21	35.24	31.80	34.13	35.33	33.76	34.00	36.56
SSIM	0.3141	0.9429	0.9525	0.9587	0.9910	0.9610	0.8810	0.9733	0.9621	0.9805	0.9880	0.9937
SAM	1.2821	0.0726	0.0918	0.0831	0.0548	0.0420	0.0375	0.0730	0.0386	0.0777	0.0632	0.0401
ERGAS	1345.32	79.99	82.50	55.06	63.73	32.96	66.87	98.31	40.58	110.29	76.95	43.24

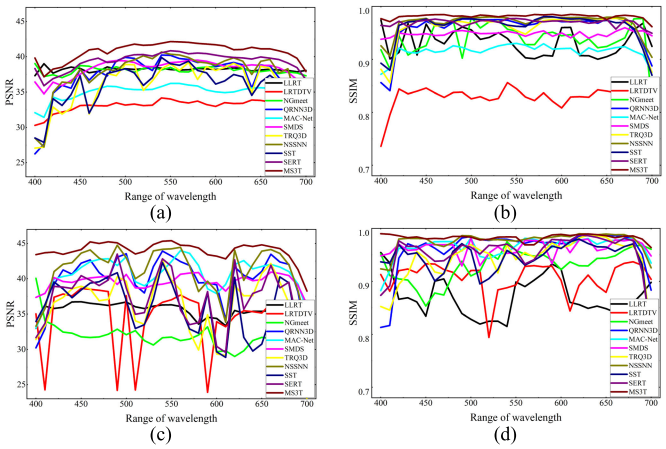


Fig. 8. PSNR and SSIM curves of different methods across diverse bands in the CAVE dataset. (a)–(d) PSNR and SSIM curves of Case 1.3 and Case 6.

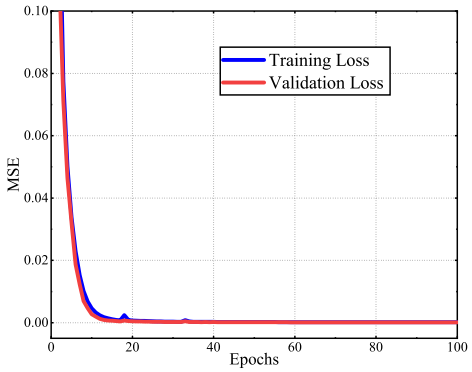


Fig. 9. Training and validation convergence curves of MS3T.

intuitively reflect the performance differences. Our method performs the best in the PSNR and SSIM metrics. However, it falls short in SAM and ERGAS metrics when compared to certain contrastive methods. This discrepancy can be attributed to the fact that PSNR and SSIM primarily focus on spatial features and pixel-level differences, whereas SAM and ERGAS consider edge and region gradient features, which are closely related to spectral content. In the NSSNN method, 3-D convolutions are the key in exploiting abundant spectral correlations in Chikusei.

3) Pavia University Dataset: Pavia University dataset has higher spectral resolution but lower spatial resolution. As demonstrated in Fig. 11, LRTDTV effectively mitigates most of the noise, however, the introduction of TV prior results in some smoothing of details. Furthermore, the enhanced constraints of LLRT and NGmeet lead to excessive smoothing and blurring. The three transformer-based methods, benefiting from the attention mechanism, adaptively focus on regions of interest, yet still introduce varying degrees of blurriness. Conversely, CNN-based QRNN3D, NSSNN, and MAC-Net perform better, with minimal loss of details. SMDS seems to lack iterative optimization for remote sensing datasets, resulting in mediocre performance. MS3T, empowered by its multiscale structure in spatial attention, maximally preserves details while removing noise. The quantitative metrics in Table IV attest to the generalization and effectiveness of MS3T.

D. Real Experiments

In this section, denoising experiments are conducted on three real-world datasets to validate the robustness of each method in complex real-world scenarios. Given the absence of clean reference images in real data, we only perform a qualitative analysis.

1) Urban Dataset: As shown in Fig. 12, the 105th band of the Urban dataset is severely corrupted by hybrid noise, leading to the complete loss of spatial details. Three model-based methods exhibit suboptimal performance owing to their limited utilization of spectral domain information for restoration. Among the learning-based methods, SMDS loses image textures during the denoising process, while QRNN3D, TRQ3D, and NSSNN remove a significant portion of Gaussian noise but retain structured noise. By incorporating low-rank memory units, SERT effectively accomplishes noise removal. Nevertheless, it introduces a certain degree of image blurring. Despite

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TABLE IV
COMPARISON OF DENOISING METRICS FOR DIFFERENT METHODS ON CASE 6 OF PAVIA UNIVERSITY DATASET

Index	Noisy	LLRT	LRTDTV	NGmeet	QRNN3D	MAC-Net	SMDS	TRQ3D	NSSNN	SST	SERT	MS3T
PSNR	18.22	32.83	36.30	35.95	37.44	38.24	31.51	36.50	38.30	37.05	37.24	39.47
SSIM	0.3219	0.9310	0.9579	0.9536	0.9922	0.9610	0.9035	0.9860	0.9808	0.9908	0.9882	0.9949
SAM	1.2453	0.0466	0.0374	0.0347	0.0452	0.0432	0.0385	0.0559	0.0451	0.0520	0.0606	0.0361
ERGAS	1310.47	74.10	47.65	48.00	44.16	42.06	81.93	55.20	43.99	50.48	53.88	34.15

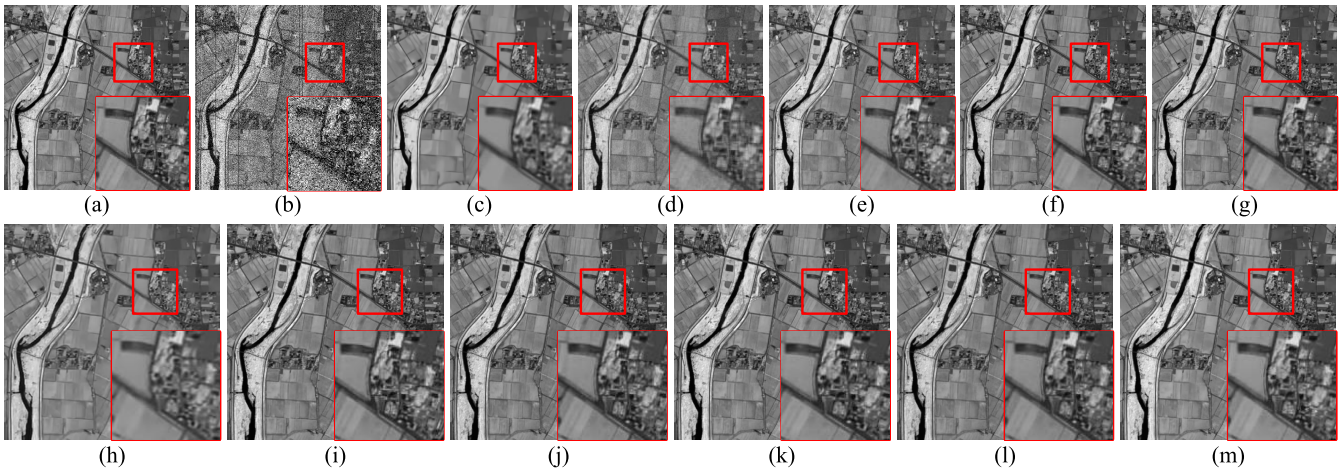


Fig. 10. Comparison of denoising results of various methods in the Chikusei dataset under Case 6, grayscale display with band 100. (a) Original image, (b) noisy image, (c) LLRT, (d) LRTDTV, (e) NGmeet, (f) QRNN3D, (g) MAC-Net, (h) SMDS, (i) TRQ3D, (j) NSSNN, (k) SST, (l) SERT, and (m) MS3T.

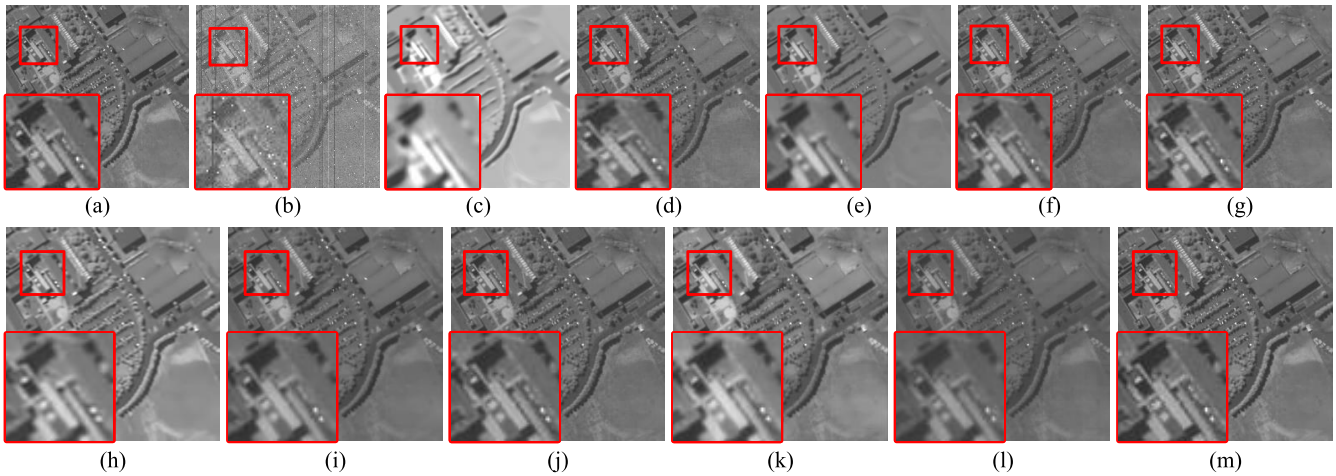


Fig. 11. Comparison of denoising results of various methods in the Pavia University dataset under Case 6, grayscale display with band 71. (a) Original image, (b) noisy image, (c) LLRT, (d) LRTDTV, (e) NGmeet, (f) QRNN3D, (g) MAC-Net, (h) SMDS, (i) TRQ3D, (j) NSSNN, (k) SST, (l) SERT, and (m) MS3T.

attaining a good equilibrium between noise suppression and feature preservation, MAC-Net suffers from spectral distortion. Under the dual constraints of spatial and spectral domains, both SST and our method demonstrate relatively optimal performance.

2) Cuprite Dataset: The 223rd band of the Cuprite dataset primarily contains Gaussian noise and impulse noise, as illustrated in Fig. 13. During the denoising process, it is susceptible to interference, leading to the loss of image details. Therefore, achieving a balance between noise removal and detail preservation poses a challenge for all methods. While LRTDTV, MAC-Net, and TRQ3D preserve textures in their results, they still exhibit residual Gaussian noise. In contrast, SST successfully removes noise but loses edge information. LLRT, NGmeet, and SMDS demonstrate varying degrees of blurring, requiring more fine-tuned parameter adjustments. SERT retains stripe noise after restoration. The results of QRNN3D, NSSNN, and our method display clear textures with almost no residual noise.

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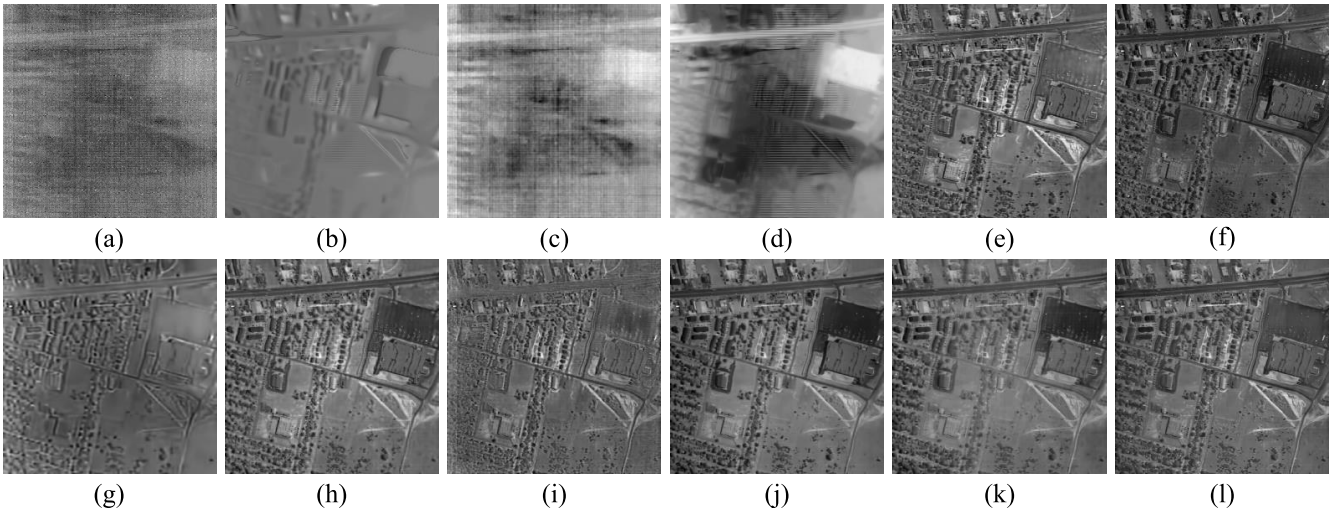


Fig. 12. Comparison of denoising results of various methods in the Urban dataset under real experiment, grayscale display with band 105. (a) Noisy image, (b) LLRT, (c) LRTDTV, (d) NGmeet, (e) QRNN3D, (f) MAC-Net, (g) SMDS, (h) TRQ3D, (i) NSSNN, (j) SST, (k) SERT, and (l) MS3T.

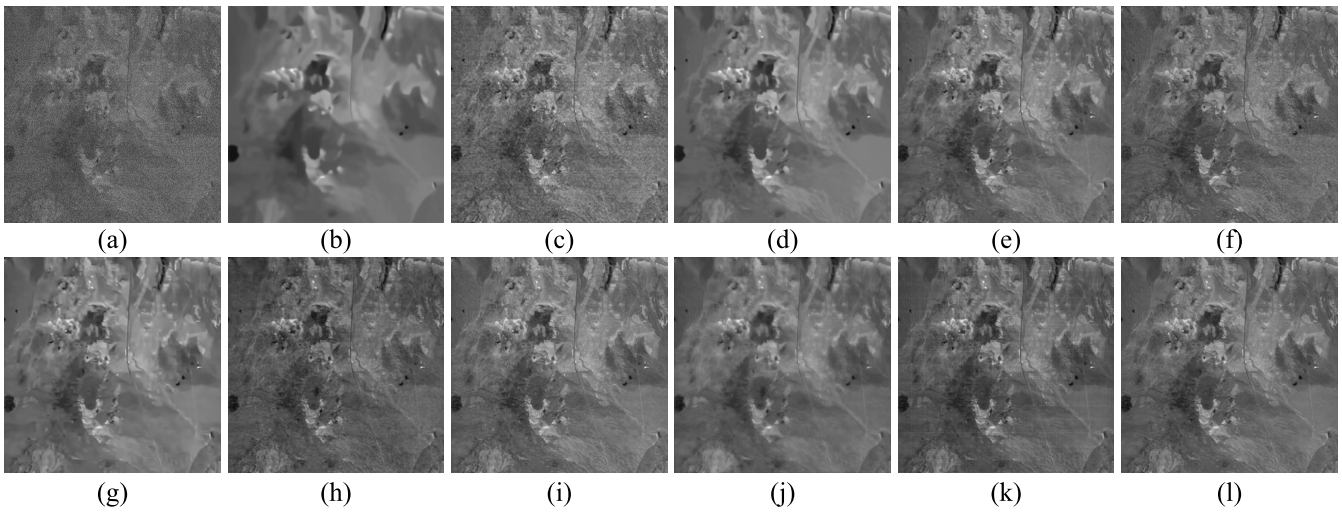


Fig. 13. Comparison of denoising results of various methods in the Cuprite dataset under real experiment, grayscale display with band 223. (a) Noisy image, (b) LLRT, (c) LRTDTV, (d) NGmeet, (e) QRNN3D, (f) MAC-Net, (g) SMDS, (h) TRQ3D, (i) NSSNN, (j) SST, (k) SERT, and (l) MS3T.

3) Indian Pines Dataset: Gaussian and stripe noise severely corrupt the first few bands of the Indian Pines dataset. Fig. 14 presents the original and denoising results of Band 3. It can be observed that LLRT, NGmeet, QRNN3D, MAC-Net, and TRQ3D effectively remove a significant portion of Gaussian and impulse noise but exhibit average performance in removing stripe noise. Despite removing most noise, SMDS leads to over-smoothing results. The remaining methods demonstrate visually similar results with no noticeable residual noise.

We conduct classification experiments on the denoised results to further validate the effectiveness of the proposed method. Employing support vector machine (SVM) on the Indian Pines dataset with 16 ground-truth labels, we assess classification accuracy through overall accuracy (OA) and kappa coefficient, as presented in Table V. Additionally, the visual results are depicted in Fig. 15, where our results exhibit more competitive performance compared to other contrastive methods.

4) River Dataset: The River dataset has certain bands severely degraded due to mixed noise such as stripe and Gaussian noise. The visual comparison of denoising experiments is illustrated in Fig. 16. In the experiment, most methods are ineffective in completely removing all noise. NSSNN removes a large amount of noise but introduces significant artifacts and smearing. TRQ3D, SST, and SERT smooth out image textures during noise removal, resulting in substantial loss of details and some spectral distortion. While our method retains some stripe noise, it achieves relatively cleaner noise removal while preserving texture information.

E. Ablation Study

In this section, ablation experiments are performed on various modules and key parameters of MS3T to investigate their impact on the model. The experimental condition is Case 1.3 of the CAVE.

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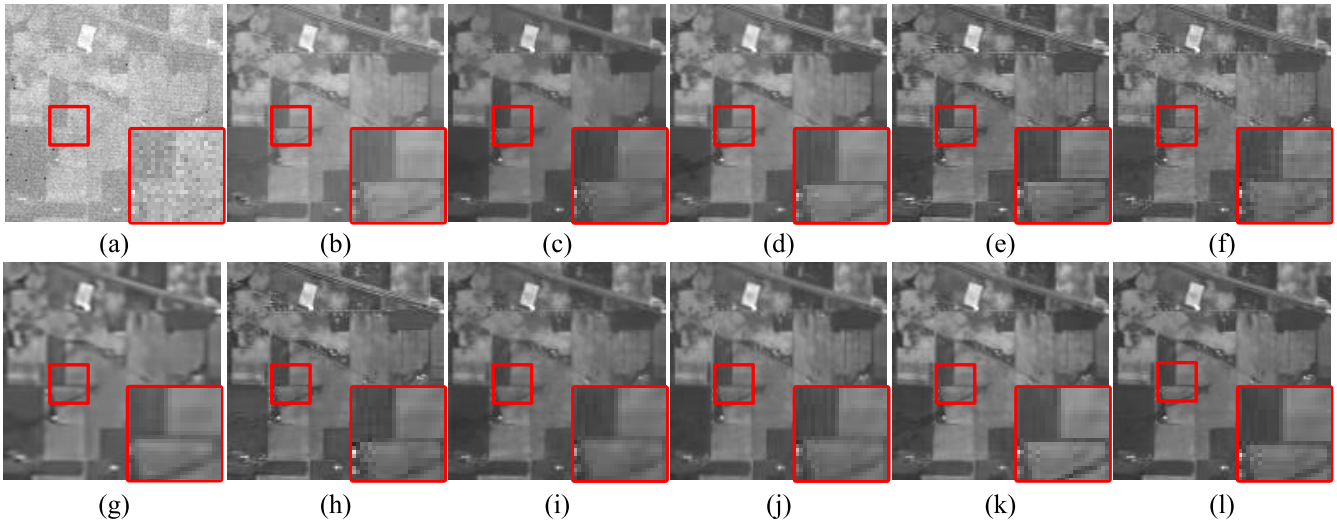


Fig. 14. Comparison of denoising results of various methods in the Indian Pines dataset under real experiment, grayscale display with band 3. (a) Noisy image, (b) LLRT, (c) LRTDTV, (d) NGmeet, (e) QRNN3D, (f) MAC-Net, (g) SMDS, (h) TRQ3D, (i) NSSNN, (j) SST, (k) SERT, and (l) MS3T.

TABLE V
COMPARISON OF CLASSIFICATION METRICS FOR DIFFERENT METHODS ON THE INDIAN PINES DATASET

Index	Original	LLRT	LRTDTV	NGmeet	QRNN3D	MAC-Net	SMDS	TRQ3D	NSSNN	SST	SERT	MS3T
OA	53.02%	66.10%	61.19%	69.64%	77.43%	72.34%	73.23%	63.04%	80.91%	69.92%	72.69%	84.41%
Kappa	0.6506	0.7536	0.7169	0.7777	0.8347	0.7746	0.8044	0.7297	0.8602	0.7796	0.8001	0.8859

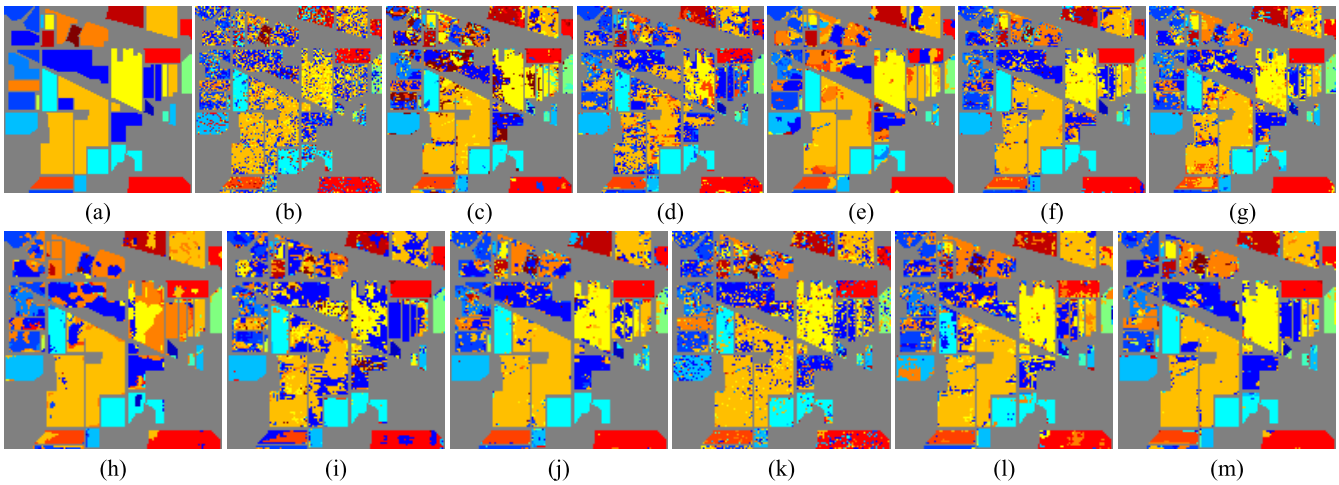


Fig. 15. Classification results of the Indian Pines dataset. (a) Ground-truth, (b) original, (c) LLRT, (d) LRTDTV, (e) NGmeet, (f) QRNN3D, (g) MAC-Net, (h) SMDS, (i) TRQ3D, (j) NSSNN, (k) SST, (l) SERT, and (m) MS3T.

1) Impact of Modules: We utilize a transformer-based residual network as our baseline model, with the self-attention mechanism of the transformer adopting a similar parallel dual-attention structure as our model, including window multi-head self-attention (W-MSA) and global spectral self-attention (G-MSA). During the ablation experiments, W-MSA and G-MSA modules are replaced with the proposed MMSA and SMSA modules. The ablation experiment results are shown in Table VI, where MMSA better simulates local and non-local similarities in space, aiding the capture of spatial consistency for more effective denoising. Furthermore, the introduction of SMSA reduces the sensitivity of attention calculations to noise.

2) Impact of Modules Arrangement Structures: Currently, many residual transformer-based methods [43], [44], [83] adopt parallel spatial–spectral attention. Given the excellent performance of these works, we choose to follow the parallel structure and conduct ablation experiments on parallel and serial structures for quantitative comparison. We also adjust the order of MMSA and SMSA in the serial structure to exclude the influence. As shown in Table VII, the indicators of the parallel structure are slightly higher than those of the two serial structures.

3) Impact of the Network Depth: In residual networks, both excessively large and small depths can lead to drawbacks.

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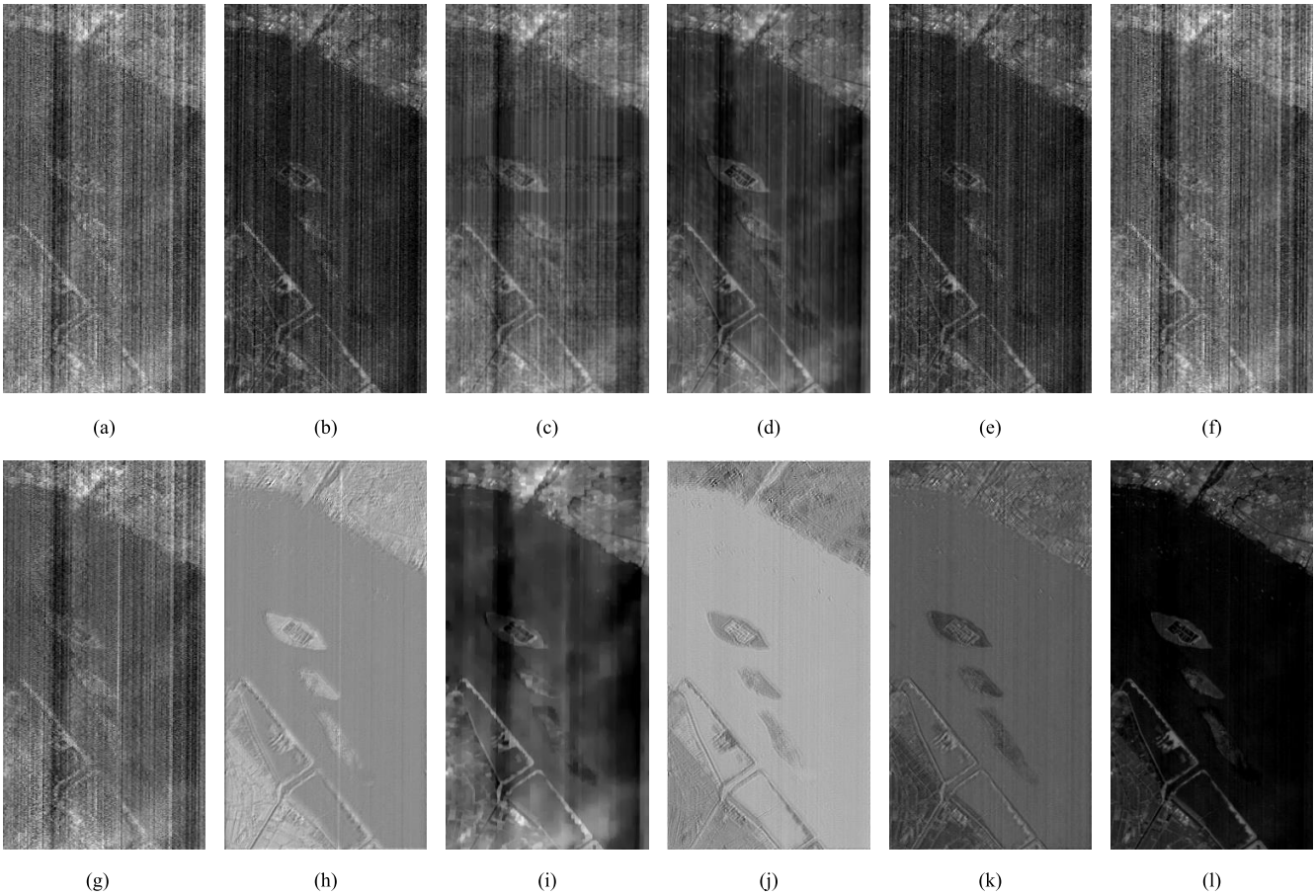


Fig. 16. Comparison of denoising results of various methods in the River dataset under real experiment, grayscale display with band 100. (a) Noisy image, (b) LLRT, (c) LRTDTV, (d) NGmeet, (e) QRNN3D, (f) MAC-Net, (g) SMDS, (h) TRQ3D, (i) NSSNN, (j) SST, (k) SERT, and (l) MS3T.

TABLE VI

QUANTITATIVE EVALUATION OF THE IMPACT OF EACH MODULE IN CASE 1.3 OF THE CAVE DATASET

Model	PSNR	SSIM	SAM	ERGAS
Baseline	39.81	0.9938	0.1824	53.58
Baseline + MMSA	40.13	0.9977	0.1571	47.02
Baseline + SMSA	40.49	0.9984	0.1580	45.61
Baseline + MMSA + SMSA	40.75	0.9985	0.1414	43.98

TABLE VII

QUANTITATIVE EVALUATION RESULTS OF DIFFERENT ARRANGEMENT STRUCTURES OF MODULES IN CASE 1.3

Structure		PSNR	SSIM	SAM	ERGAS
In serial	MMSA-SMSA	40.37	0.9981	0.1556	46.22
	SMSA-MMSA	40.51	0.9984	0.1534	45.09
In parallel	MMSA-SMSA	40.75	0.9985	0.1414	43.98

Excessive depth results in diminishing gradients and weakened weights, leading to performance degradation. Insufficient depth fails to capture complex features, limiting the expressive capacity. Therefore, an appropriate depth is essential, enabling efficient training while adequately capturing data characteristics. Following empirical findings of prior work [83], we set TRL numbers and MSTB numbers in each TRL within a certain range to quantitatively analyze the impact of network depth on denoising performance. As shown in Table VIII, the best performance is achieved when the numbers of TRL and MSTB are both six.

TABLE VIII

QUANTITATIVE EVALUATION OF THE IMPACT OF NETWORK DEPTH IN CASE 1.3 OF THE CAVE DATASET

TRLs	MSTBs	PSNR	SSIM	SAM	ERGAS	Param(M)
3	6	38.01	0.9919	0.1841	54.15	1.55
4		39.39	0.9947	0.1620	51.92	2.03
6		40.75	0.9985	0.1414	43.98	2.98
8		40.73	0.9983	0.1495	44.33	3.93
6	3	38.75	0.9892	0.1736	53.46	1.77
	4	39.81	0.9953	0.1558	51.34	2.17
	6	40.75	0.9985	0.1414	43.98	2.98
	8	40.64	0.9980	0.1462	45.29	3.78

4) Sensitivity Analysis of Block Size: In MMSA, the size of the blocks has an impact on the ability to extract spatial information. When the block size is small, the perception of the global context is limited. However, as the block size continues to increase, the model cannot fully utilize local spatial similarity, causing the loss of partial details. We conduct the

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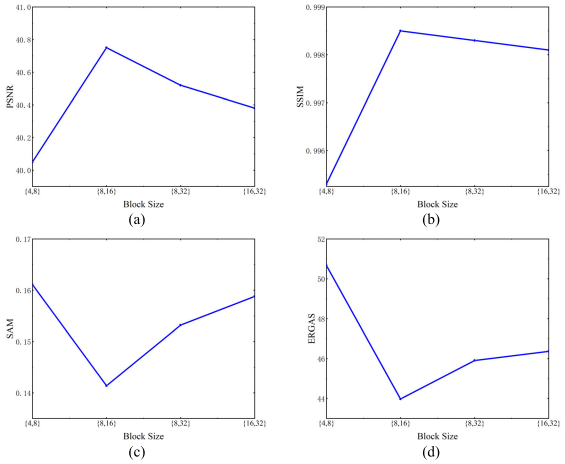


Fig. 17. Sensitivity analysis of block size in Case 1.3. (a) PSNR, (b) SSIM, (c) SAM, and (d) ERGAS.

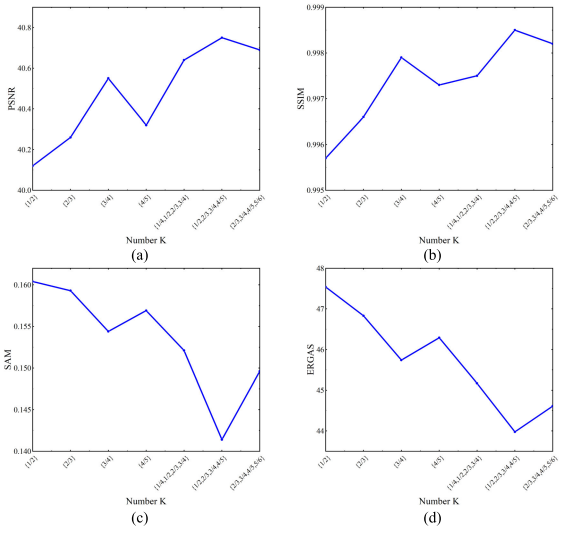


Fig. 18. Sensitivity analysis of number k in Case 1.3. (a) PSNR, (b) SSIM, (c) SAM, and (d) ERGAS.

experimental analysis with various combinations and obtain the best performance at block sizes of {8, 16}, as illustrated in Fig. 17.

5) Sensitivity Analysis of Number k: We conduct experimental analysis on the selection of the key value k in the top-k operator, as shown in Fig. 18. Initially, we set a single fixed value to filter and extract the crucial information. It is observed that, when k is too small, the attention matrix becomes excessively sparse, resulting in significant information loss and limited model performance. Conversely, when k is too large, it fails to filter out high-frequency noise and interference effectively. Moreover, the performance of the model is significantly influenced by a single k value due to variations in image scenes. Therefore, we opt to set multiple k values within a range to adaptively retain the regions of interest while weakening the weight of irrelevant information. Ultimately, we select {1/2, 2/3, 3/4, 4/5} as the k values for this model based on their optimal performance metrics.

6) Sensitivity Analysis of Feature Channel C: Selecting an appropriate number of feature channels before deep feature extraction is crucial [20]. Based on previous work [83],

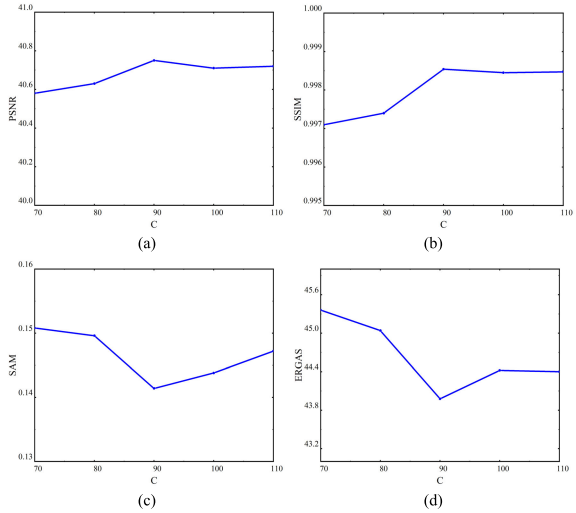


Fig. 19. Sensitivity analysis of feature channel C in Case 1.3. (a) PSNR, (b) SSIM, (c) SAM, and (d) ERGAS.

we investigate the impact of varying C values within a certain range on the performance metrics. As illustrated in Fig. 19, the performance gradually improves as C increases, eventually tending to be stable around C equals 90. When the channel is too low, the spectral information is incomplete, leading to insufficient feature extraction. On the other hand, excessively high channel introduces redundancy, computational complexity, and potential noise residual. Finally, we set C to 90 as a trade-off between information completeness and redundancy.

V. Conclusion

In this study, we present a novel HSI denoising network called MS3T, which aims to address the limitations of existing transformer-based methods in exploring the spatial–spectral characteristics of HSI and the interference of noise weights on global information interaction in attention mechanisms. Specifically, MS3T is based on a residual transformer framework, where the MMSA module is employed to establish local and non-local multiscale spatial consistency and the SMSA module effectively extracts global spectral information by masking non-correlated weights in the spectral domain. Extensive experiments conducted on natural HSIs and remote sensing HSIs verify that the proposed method surpasses other approaches in both visual perception and quantitative metrics. Additionally, the ablation studies validate the efficacy of each proposed module. In future work, we aim to tackle challenges related to the large parameter volume of transformers and the requirement for a large number of training samples.

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